How-To Library

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Enabling 360 degree View of Data using SAP BW Scanner

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Abstract

This article describes how to catalog the metadata from SAP Business Warehouse (SAP BW) systems in Enterprise Data Catalog.

Supported Versions

- Enterprise Data Catalog 10.4.0

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Overview

When using ERP systems such as SAP, business users spend a great amount of time to analyze data impact, data usage in other systems, and relationships between multiple systems.

SAP business users encounter the following challenges in the SAP BW systems:

- Data model complexity
- Configuration
- Data volume
- User interface

SAP business users can use the SAP BW scanner to extract metadata from SAP BW systems and catalog metadata in Enterprise Data Catalog. For business users, the SAP BW scanner eliminates the need to understand the complexities in the metadata objects and extracts the metadata. The extracted metadata is cataloged in Enterprise Data Catalog to enable a 360 degree view of data.

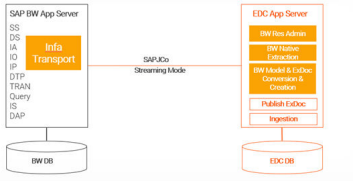
Implementing a 360 Degree View

To implement a 360 degree view of data:

- Create functional modules in the SAP BW system for metadata extraction. Enterprise Data Catalog integrates with the SAP BW system using the SAP Java Connector (SAP JCo) libraries to extract the required metadata.
- Enable functional modules to handle the memory issues by segregating metadata based on specific dates.
- Enable business users to specify the type of metadata to extract, and extract metadata based on the query execution date, object modified date, or perform a full extraction.

Architecture of SAP BW Scanner

The following image shows the architecture of the SAP BW scanner:



The SAP BW scanner extracts metadata objects such as Info Area, Data Source, Info Objects, Info Object Catalog, Source System, Characteristics, Key Figures, Dimension, Navigational Attributes, Infoset, DSO, Info Cube, Multi Provider, Transformation, Transformation Rule, Query, Query Element or Field, Workbook, Advanced DSO, Composite Provider, and DTP.

Advantages of SAB BW Scanner

The following are the advantages of using the SAP BW scanner:

- Enables the business users to understand the data flow within and across the BW system.
- Discovers the hidden, valid, and useful patterns in the metadata data sets.
- Enables the business users to derive insights that can used in marketing and scientific discovery.

Handling Complex Metadata

The SAP BW scanner addresses the following problems related to complex metadata:

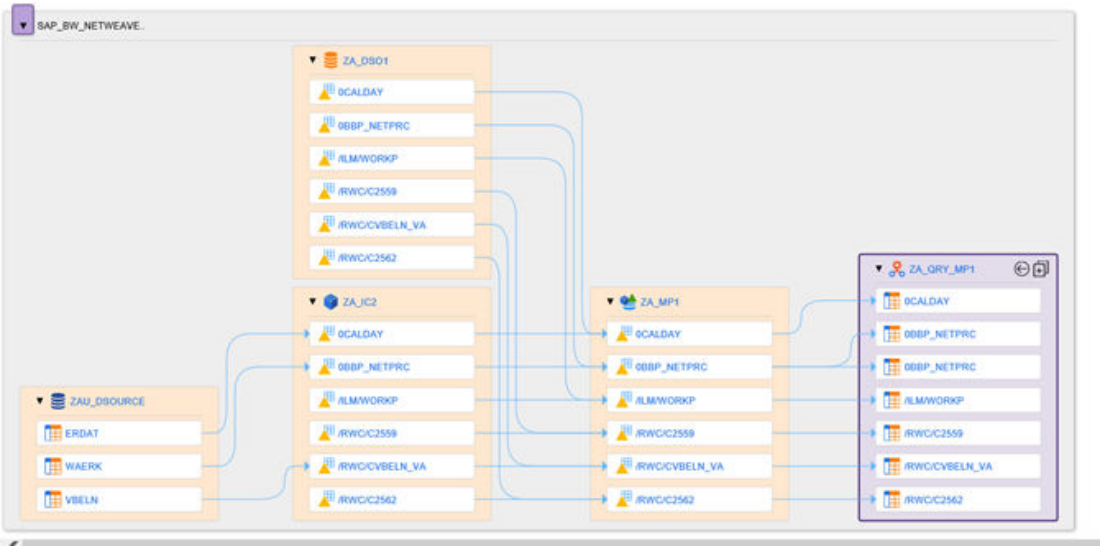
- The queries and the associated formulae are not visible in the SAP GUI client. Therefore, the business users need to traverse multiple applications to understand the query data flow.
- The transformation rules and the composite data providers created on the SAP HANA tables or views are complex.

The SAP BW scanner extracts and displays the metadata objects related to query data flows, composite data providers, and transformation rules in the Catalog.

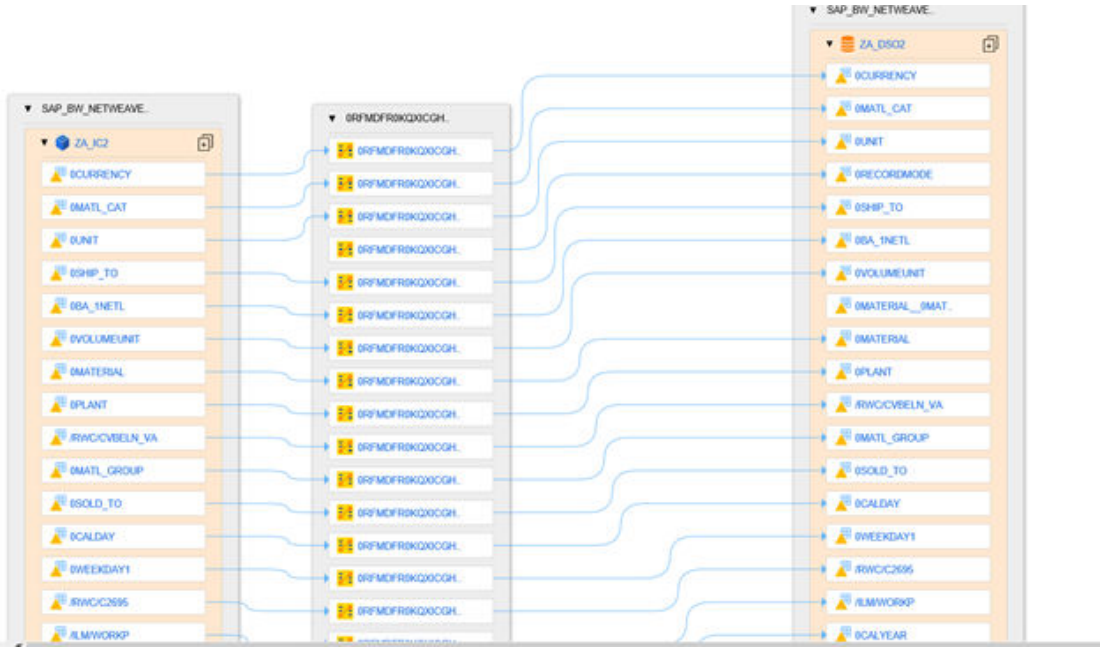
The following image shows the data flow diagram of the SAP BW system displayed in the Catalog:



The following image shows the query lineage extracted by the SAP BW scanner:



The following image shows the transformation lineage extracted by the SAP BW scanner:



Business users can use the information in the Catalog to understand the logic implemented in the SAP BW system using the SAP BW scanner.

Author

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